



Invited Review

The forgotten sons: Warehousing systems for brick-and-mortar retail chains

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ABSTRACT

Warehouses are an inevitable component in any supply chain and a vividly investigated object of research. Much attention, however, is absorbed by warehousing systems dedicated to the special needs of online retailers in the business-to-consumer segment. Due to the ever increasing sales volumes of e-commerce this focus seems self-evident, but a much larger fraction of retail sales are still realized by traditional brick-and-mortar stores. The special needs of warehouses servicing these stores are focused in this paper. While e-commerce warehouses face low-volume-high-mix picking orders, because private households tend to order just a few pieces per order from a large assortment, distribution centers of retail chains rather have to process high-volume-low-mix orders. We elaborate the basic requirements within both business segments and identify suited warehousing systems for brick-and-mortar stores (e.g., fully-automated case picking). The setup of each identified warehousing system is described, elementary decision problems are discussed, and the existing literature is surveyed. Furthermore, we identify future research needs.

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1. Introduction

Warehousing, i.e., the intermediate storage of goods between stages in a supply chain (Bartholdi & Hackman, 2016), and its basic functions, i.e., receiving, storage, order picking, and shipping (Gu, Goetschalckx, & McGinnis, 2007), are inevitable in any supply chain. Therefore, it is not surprising that a vast body of literature on this topic has accumulated over the past decades (see the most recent surveys of Azadeh, de Koster, & Roy, 2019; De Koster, Le-Duc, & Roodbergen, 2007; Gu et al., 2007; Gu, Goetschalckx, & McGinnis, 2010; Van Gils, Ramaekers, Caris, & De Koster, 2018). A large part of this research effort – especially in recent years – is absorbed by warehousing systems dedicated to the special requirements of online retailers mainly in the business-to-consumer (B2C) segment (e.g., see the large body of literature reviewed in the survey of Boysen, De Koster, and Weidinger (2019b) on e-commerce warehouses). Due to the ever increasing sales volumes of e-commerce (Statista, 2018a) this focus seems self-evident, but a much larger fraction of retail sales are still realized by tradi-

tional brick-and-mortar stores. In spite of impressive growth rates, according to a recent estimate, the e-commerce share of total global retail sales will only reach 13.7% in 2019 (Statista, 2018b). The warehousing literature does not mirror this relation; literature specifically dedicated to warehouses servicing brick-and-mortar retail chains is (comparatively) rare. This survey paper intends to extenuate this bias. We review the scientific literature on warehouse operations fulfilling the demands of store-based retail chains and identify important future research needs.

The need for differently structured warehouses when either participating in e-commerce or servicing brick-and-mortar stores is mainly caused by deviating order characteristics. While online retailers face low-volume-high-mix picking orders, distribution centers for retail chains rather have to process high-volume-low-mix picking orders. In the B2C segment of online retailing, private customer households tend to order just a small number of products per order from a broad assortment, whereas brick-and-mortar stores offer a much smaller assortment in their restricted shop space, but typically order multiple pieces of each stock keeping unit (SKU). We detail the main requirements for warehouses servicing brick-and-mortar stores (in comparison to e-commerce) in Section 2, where we also identify suited warehousing systems that are able to satisfy (most of) the identified requirements. This defines the scope of our survey. Each of the identified warehousing systems (namely fully-automated case picking, pick-and-pass

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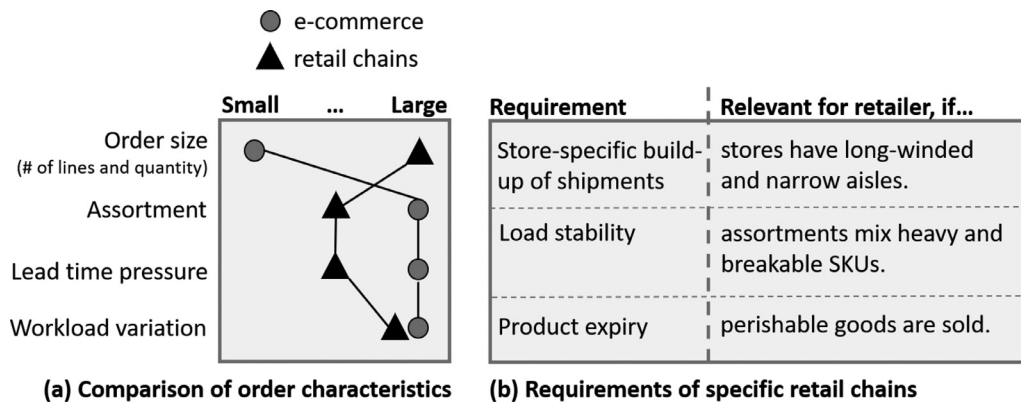


Fig. 1. Order characteristics of e-commerce vs. brick-and-mortar retail chains.

systems, crane-supplied pick faces, bulk picking, put systems, and picker-to-parts systems with vehicle support) is dedicated a separate section, where we describe the respective system, discuss the fit for the basic requirements, elaborate the main decision problems to be solved when setting up and operating such a specific warehousing system, and survey the existing literature. Finally, Section 9 concludes the paper.

Due to the great practical relevance of order fulfillment in the recent years, it is our impression that warehousing research revitalizes and many new researchers (are about to) enter this field. Therefore, our descriptions are intended general enough, so that OR researchers and practitioners not familiar with warehousing yet can understand the basic system setups, processes, decision problems, and future research needs. However, we hope that also established warehousing researchers receive a compact and concise (re-)introduction of warehousing systems that are not in the lime-light of the warehousing literature.

2. Scope of survey: requirements and suited warehousing systems

Whether a warehousing system fits a specific application in an industry sector is mainly determined by the characteristics of customer orders that have to be processed. Note that, from the warehouse perspective, a customer does not need to be a final customer in the B2C sense, but can also be a sales outlet or a B2B customer. The specific order characteristics or requirements for warehousing systems that either service B2C online sales or brick-and-mortar stores are compared in Fig. 1 and elaborated in the following:

- (i) Order size: the average order demand at German Amazon warehouses only amounts to 1.6 pieces per order (Boysen, Stephan, & Weidinger, 2019c). Typically, online retailers face small-sized orders, which consist of just a few order lines (often only a single line) each demanding one or just a few pieces. Brick-and-mortar stores, instead, have continuous demands triggered by customer sales. This allows to bundle demands to larger orders with multiple order lines and multiple pieces per line, so that transportation costs are reduced and order handling within stores is facilitated (Sternbeck & Kuhn, 2014).
- (ii) Assortment: costs of storage space in a remote warehouse of an online retailer are much lower than those for the scarce and costly shelf space in retail stores located in urban areas. Therefore, online retailers can afford a much larger assortment and niche products account for a much larger proportion of sales in e-commerce than they do in brick-and-mortar stores. Within both segments, however, the variety is huge. The range in retail business is from hundreds of

SKUs in small mom-and-pop stores, via tens of thousands in a typical supermarket, up to hundreds of thousands in some very large department stores (Kök, Fisher, & Vaidyanathan, 2008). In many businesses, however, the assortments in e-commerce tend to be larger; this phenomenon is also known under the term “the long tail” of e-commerce (Brynjolfsson, Hu, & Smith, 2003).

- (iii) Lead time pressure: next-day or even same-day deliveries are an elementary promise of most online retailers (e.g. Yaman, Karasan, & Kara, 2012), so that the warehousing processes in e-commerce face extremely tight delivery schedules. Due to the demand bundling elaborated above most retail stores are serviced under a fixed delivery pattern, which is cyclically repeated each week (Holzapfel, Hübner, Kuhn, & Sternbeck, 2016). These fixed schedules ease planning and reduce time pressure. However, novel services in omni-channel retailing, i.e., click-and-collect where customers order online and pick up within a store (Hübner, Holzapfel, & Kuhn, 2016a; Hübner, Kuhn, & Wollenburg, 2016b), and the fierce competition of online retailers lead to a considerable, yet still lower time pressure for warehouses of retail chains.
- (iv) Workload variation: sales hugely vary over time both in online and offline distribution channels. Over the week, most warehouses of online retailers have their peak workload on Mondays when processing the orders of the weekend (Poggi, Carrera, Gavalda, Ayguadé, & Torres, 2014). Offline retail chains have to ensure that shelves are filled for the peak sales on the weekend and refilled afterwards, so that most warehouses of brick-and-mortar stores have their peak day on Fridays (Li & Lim, 2018) and Mondays. Moreover, sales also vary over the year where, for instance, much higher sales occur during end-of-season sales (Chevalier, Kashyap, & Rossi, 2003; Kembro, Norrman, & Eriksson, 2018). Thus, warehousing systems in both domains have to flexibly react on varying workloads, which is harder if a warehousing system relies on fixed installed hardware, such as conveyors, lifts, shuttles, or cranes.

Clearly, the importance of these general characteristics for warehouses servicing brick-and-mortar stores varies, e.g., with the business strategy and the offered products. Many luxury stores, for instance, have rather erratic sales for individual products, so that their order profiles rather resemble the low-volume-high-mix orders of online retailing. Other stores, e.g., tobacco shops, have a rather stable demand over the year, and warehouses servicing drugstores and hospitals with urgent pharmaceuticals operate under great time pressure. We, however, exclude these special cases and assume that the four characteristics (i)–(iv) are important when selecting a warehousing system suited for brick-and-mortar

retail chains. In addition to these core characteristics derived from the general order characteristics faced by retail chains (see Fig. 1(a)), we have three additional requirements which are only relevant for specific store-based retailers (see Fig. 1(b)).

- (v) *Store-specific build-up*: due to the high real estate prices in most urban areas stores are often small with densely arranged shelves, so that maneuvering newly arrived goods through narrow aisles is time consuming and obstructs customers. In such a setting, a warehousing system that is able to build up the shipments, e.g., within roll cages, in the order defined by the layout of the respective store can considerably facilitate the shelf filling process. To realize this requirement, a warehousing system has to be able to convey the demanded pieces in a sequence suiting the respective store layout, such that the build-up process can add one piece after another onto a roll cage. If store layouts are standardized, e.g., in a centrally controlled retail chain, the storage assignment of SKUs in the warehouse can directly mirror the shelf arrangement in the standardized store, which Hübner, Holzapfel, and Kuhn (2015) call *outlet picking*. With properly made up shipments, the sales staff do not need to zigzag through the store but can conveniently follow the store layout. This is especially valuable if the sales staff, e.g., in an urban area, causes high wage costs compared to logistics workers, e.g., of a remote warehouse in a rural area. We call this requirement a *store-specific build-up* of shipments.
- (vi) *Load stability*: some retailers have assortments containing a mix of heavy and fragile SKUs, so that within a shipment (e.g., in a roll cage) heavy products may not be placed on top of fragile SKUs (Dyckhoff, 1990; Žulj, Glock, Grosse, & Schneider, 2018). This requirement is, for instance, relevant for food retailers handling heavy beverage crates and fragile egg cartons. A warehousing system suited for this requirement has to ensure the proper sequence of SKUs, so that heavy products to be placed on the bottom of a roll cage are presented before fragile SKUs. Note that, from the warehouse perspective, the main difference between stability issues and store-specific build-ups, is that the former allows for a sorting of products (from heavy and stable to lightweight and fragile) that is identical for all stores. This allows, for instance, to derive a storage assignment ordered by stability aspects. A store-specific build-up may vary from store to store, so that no unique sorting exists. Further note that for e-commerce and its small-sized orders, stability might be an issue in some branches, e.g., online food retailing, but it is generally of less importance.
- (vii) *Product expiry*: if perishable goods (e.g., flowers, foodstuffs, or pharmaceuticals) are handled by a retailer, a suited warehousing system has to be able to prioritize those pieces of each SKU with the closest date of expiry. See (Akkerman, Farahani, & Grunow, 2010) for a general survey on food quality. Note that this requirement is, typically, realizable with any refrigerated warehousing system, as long as the underlying information system is able to identify the prioritized pieces. Once this is possible, the warehousing system only has to be able to retrieve the identified products. A warehousing component that is hardly suited at least in deep-freeze warehouses is a bag or pouch sorting system discussed in Section 6. Further note that, obviously, e-commerce also has to consider product expiry when selling the aforementioned products. However, compared to other e-commerce branches the market share of online food retailers is still small and most sales are still made via brick-and-mortar stores (Kembro et al., 2018). Thus, product expiry is less of an issue in most e-commerce warehouses.

Note that there are some further peculiarities of brick-and-mortar retail chains that influence warehouse operations, although in a more indirect manner as those listed before. Brick-and-mortar retailers are much less affected by returned products compared to e-commerce (De Koster, de Brito, & van de Vendel, 2002). Furthermore, warehouses of retail chains have access to point-of-sales data, so that they can plan incoming demands well ahead, which gives them much more control over transportation. Finally, especially perishable goods have to be moved quickly from suppliers to sales outlets, so that long-term storage in warehouses is impossible for these goods. Therefore, warehouse operations of many retail chains are closely coupled with cross-docking (e.g., Boysen & Fliedner, 2010). We discuss these influencing aspects and their impact on warehousing in more detail in Section 9.

System selection: this directly leads us to the problem how to identify whether a specific warehousing system fulfills our requirements (i) to (vii) and should, thus, be integrated into this survey. A warehousing system consists of *hardware*, which can be further subdivided into storage systems (e.g., a rack), material handling systems (e.g., a conveyor belt) and picking tools (e.g., a pick-by-voice solution), and *processes* defining the work flow along the applied hardware elements (e.g., sequential zone picking) (Boysen et al., 2019b). There exist plenty alternative warehousing systems applying varying hardware and processes; an overview is provided by the survey papers of Gu et al. (2007, 2010), De Koster et al. (2007), and Azadeh et al. (2019). Since deriving objective selection criteria seems barely possible, we follow the general selection procedure of Boysen et al. (2019b), which they apply for identifying warehousing systems suited for e-commerce.

First, we preselected the list of warehousing systems depicted in Fig. 2 (ordered according to their level of automation). This list excludes all warehousing systems dedicated to online retailing as defined by Boysen et al. (2019b) and is based on numerous warehouse visits, interviews, and the websites of manufacturers of warehousing solutions. Specifically, 15 Dutch warehousing managers and experts have been questioned in semi-structured interviews after warehouse tours. Then, the authors rated whether or not the respective warehousing system fulfills requirements (i)–(vii) on a binary scale and discussed the plausibility of the outcome with yet 15 additional warehousing experts (scientists and consultants). The result is depicted in Fig. 2 where a system fitting the respective requirement is indicated by the check icon. Note that our requirements should not be interpreted as must-have requirements, but rather as design-goals which a specific warehousing system should enable in order to be well-suited for brick-and-mortar retailers. Based on these results we review all warehousing systems fulfilling at least three out of four among the core requirements (i)–(iv) generally relevant for all retailers. Note that the identified warehousing systems vary in the number of installations. The predominant warehousing system, said to be followed by 80% of all Western European warehouses (De Koster et al., 2007; Napolitano, 2012), is still the picker-to-parts system which, due to the larger orders, requires order picking vehicle support in store-based retail business. The other five types are more recent systems that try to improve over the established picker-to-parts processes by applying new processes and/or novel hardware.

Beyond our scope: Due to our selection procedure, we exclude the following warehousing systems from our survey:

- *A-Frame* systems consist of a set of successive vertical channels filled with stockpiles of SKUs (see also Boywitz, Schwerdfeger, & Boysen, 2019). The channels are arranged along two opposite frames, which have their top ends tilted toward each other like a pitched roof. A conveyor belt runs between the two frames, so that from the side view the system looks like an “A”. At the bottom of each channel an automated dispenser pushes

		suitability for general order characteristics				branch specific requirements		
		(i) large orders	(ii) medium assortments	(iii) medium time pressure	(iv) varying workload	(v) store-specific build-up	(vi) load stability	(vii) product expiry
level of automation	A-Frame systems			☑			☑	☑
	Compact storage systems	☑	☑				☑	
	Fully-automated case picking	☑	☑	☑		☑	☑	☑
	Pick-and-pass systems	☑	☑	☑	☑		☑	☑
	Crane-supplied pick face	☑	☑	☑				☑
	Bulk picking and store consolidation	☑	☑	☑			☑	☑
	Put systems		☑	☑	☑			☑
	Picker-to-parts with vehicle support	☑	☑	☑	☑	☑	☑	☑
	Picker-to-parts without vehicle support		☑		☑	☑	☑	☑

Fig. 2. Overview of warehousing systems suited brick-and-mortar retail chains.

one or more bottommost pieces toward the conveyor whenever the respective products are required by an order passing by on the conveyor. A-Frames are very efficient, i.e., they process up to 750,000 pieces per day at an order accuracy of more than 99.95% (Caputo & Pelagagge, 2006), but require plenty human workers for refilling the channels. This is especially true if large orders demanding plenty pieces per order line are to be processed. Furthermore, already mid-sized assortments require excessive investment into fixed installed A-Frame modules, which are, thus, hardly scalable. A-Frames are rather suited for a restricted number of the most fast-moving products of relatively small and infrangible products, e.g., pharmaceuticals and cosmetics (Bartholdi & Hackman, 2016; Pazour & Meller, 2011) and, thus, excluded from our survey.

- In the recent years, quite a few automated compact storage systems have been developed. Examples are movable rack systems (Boysen, Briskorn, & Emde, 2017), shuttle-based deep lane storage systems (Boysen, Boywitz, & Weidinger, 2018a; Zaerpour, Yu, & De Koster, 2015), the live-cube system (Zaerpour, Yu, & De Koster, 2017), and puzzle-based storage systems (Gue & Kim, 2007). A detailed overview on compact storage is provided by the survey paper of Azadeh et al. (2019). These systems use space efficiently by storing products closer together without providing a direct access to each single piece (at any time). Thus, they struggle to provide fast access times. This may especially be a problem, if a specific unit load of a SKU is to be retrieved, e.g., due to an approximating date of expiry. Furthermore, the large amount of fixed hardware makes these systems hardly scalable.
- More than 80% of all warehouses in Western Europe still follow the traditional picker-to-parts setup (De Koster et al., 2007; Napolitano, 2012). In these warehouses, SKUs are stored in low-level racks, and human order pickers collect requested pieces by traveling from shelf to shelf. Without vehicle support, i.e., a motorized order pick truck driven by a picker, it is hardly pos-

sible to handle large orders, especially if pallet picking leads to large distances to be covered and tight deadlines have to be met. Note that very small and lightweight SKUs (e.g., cosmetics) enable pickers to still handle larger store orders without vehicle support, e.g., by pushing a manual picking trolley. Nonetheless, we exclude this case from our survey. Small and lightweight SKUs are not the norm in store-based retailing and traditional picker-to-parts warehouses without motorized vehicle support have already been treated in detail by previous survey papers (De Koster et al., 2007; Gu et al., 2007; 2010; Masae, Glock, & Grosse, 2019).

- We also exclude warehousing systems dedicated to specific product types requiring special handling. In the fashion industry, for instance, there are automated sorting systems for hanging goods (Schäfer, 2018a). Here, goods-on-hangers, e.g., dresses and coats, are stored on slick rail lanes and can automatically be sorted into lanes assigned to a specific store. Another example are crane-based systems especially designed for the handling of tires (Cimcorp, 2011). We briefly come back to these special systems in Section 9.
- Finally, prototype systems that have not outgrown the evaluation phase and are yet not applied by large retail chains are also excluded. Examples are the Toru robot of Magazino (2017) or the Robo-Pick system of Schäfer (2018b), which promise fully automated order picking from shelves and bins, respectively.

Structure of each section: excluding these warehousing systems leaves behind the six warehousing systems that are discussed in the following sections (ordered by decreasing level of automation and in the same order depicted in Fig. 2). For each system we elaborate the following aspects:

- System description: we give a brief introduction of the main system elements and their role in the order fulfillment process. Note that this description focuses on outbound processes, because they are the main drivers for customer

satisfaction, costs, and planning complexity. The inbound process and the put-away of pieces into shelves receives less attention.

- (b) Suitability for warehouses servicing retail chains: given our seven requirements, we briefly summarize the fit of each system and provide examples where in the retail sector they are applied.
- (c) Decision problems: for each system, basically the following three main decision areas exist (ordered according to planning horizon) (Boysen et al., 2019b): (i) layout design planning, (ii) storage assignment, and (iii) order processing. We discuss what peculiarities have to be considered when addressing these planning tasks for the respective warehousing system.
- (d) Literature survey and future research: Finally, we also serve the main intention of a survey paper, that is we review the relevant literature and outline future research needs.

Relation to previous surveys: there already exist quite a few survey papers in the warehousing research area, so that, before going on, we have to clarify whether yet another survey can be justified. Among the existing survey papers there are the general surveys of (Van den Berg, 1999; De Koster et al., 2007; Gu et al., 2007; 2010; Rouwenhorst et al., 2000). These papers address the complete field of warehousing, but mostly focus on traditional picker-to-parts systems. Since their publication, some time has passed and warehousing systems specially suited for brick-and-mortar retail chains are, if at all, just briefly addressed. Other surveys focus on special warehousing aspects: automated storage and retrieval systems (ASRS) (Gagliardi, Renaud, & Ruiz, 2012; Roodbergen & Vis, 2009), crane scheduling in ASRS (Boysen & Stephan, 2016), robotized warehousing systems (Azadeh et al., 2019), holistic warehousing research that combines two or more planning tasks (Van Gils et al., 2018), stochastic warehousing models (Gong & De Koster, 2011), e-fulfillment in a multi-channel environment (Agatz, Fleischmann, & van Nunen, 2008), picker routing (Masae et al., 2019), and e-commerce warehouses (Boysen et al., 2019b). Thus, no existing survey is dedicated to our specific topic.

Literature retrieval: Finally, we briefly specify our database search for retrieving the papers to be reviewed (see, e.g., Hochrein and Glock (2012) for a general description of how to set up a systematic literature retrieval). As keywords specifying the business function we apply “warehousing”, “distribution center”, “order fulfillment”, and “order processing”. Furthermore, a second group of keywords, i.e., “brick-and-mortar”, “retailing”, “store chain”, “offline channel”, and “retail chain”, is applied to specify the branch of industry. Any combination of keywords from the first and second group has been applied as a query in two scholarly databases, namely Business Source Premier and Scopus. Additional queries were executed with the names of the six reviewed warehousing systems. All English-language papers published in peer-reviewed journals that have been retrieved and those cited in their reference lists (snowball approach) were checked for relevance by analyzing their abstracts. Note that we try to build up on the work of existing survey papers as far as possible to avoid repetition. Picker-to-parts systems, for instance, are surveyed in detail by Gu et al. (2007) and De Koster et al. (2007). We address picker-to-parts systems with motorized vehicle support, so that we try to avoid overlap and concentrate on papers published after these previous two surveys have appeared.

3. Fully-automated case picking

Fully-automated case picking is the only warehousing system surveyed in this paper where the complete order fulfillment process is automated. The assembly of mixed pallets or roll cages ac-

cording to the orders of brick-and-mortar stores is executed by industrial robots. Current generation industrial robots are restricted to handling stable, rectangular and not-too-small products, so that SKUs not fulfilling these requirements have to be unified into cases, e.g., a dozen of milk cartons stacked together. The basic steps of the fulfillment process are depicted in Fig. 3.

Incoming goods arriving in homogeneous unit loads (predominantly pallets) are, first, fed onto a conveyor, which delivers them toward the depalletizing step. Here, individual cases are isolated, e.g., by an industrial robot depicted in step 1 of Fig. 3. Either loose cases are directly put onto another conveyor or they are transported in trays to enable a fail-safe handling in an automated storage and retrieval system (ASRS) (Azadeh et al., 2019). Step 2 of Fig. 3 displays a lift-and-shuttle system (Carlo & Vis, 2012), but also crane-operated racks (Boysen & Stephan, 2016; Roodbergen & Vis, 2009) or carousel systems (Litvak & Vlasiov, 2010) can be applied for the intermediate storage of cases. Within the ASRS, the cases wait for the orders transmitted by brick-and-mortar stores. According to these orders, requested cases are retrieved from the ASRS (step 3 in Fig. 3) and brought into a proper sequence on another conveyor, such that stability aspects and/or a store-specific build-up can be ensured by the palletizing machine (step 4 of Fig. 3). Here, another industrial robot stacks the cases onto mixed pallets or roll cages according to a packing pattern, which is automatically generated by an optimization procedure integrated into the underlying IT system. For the palletizing step, there exist alternative technologies. Instead of lifting each case individually, alternative approaches build a layer first, which is then completely lifted by an industrial robot. Other approaches push the cases one after another onto a retractable pallet, e.g., OPM by Witron, or apply human workers for palletizing non-standard cases, e.g., the manual I-packstation of mixed palletizing (WSR BV). Once the pallets or roll cages are assembled, they are moved (e.g., via another conveyor) to the shipping area where they are loaded onto outbound trucks.

Fully-automated case picking fulfills all our seven requirements with regard to the suitability for servicing the demands of brick-and-mortar stores except for one. It applies a lot of fixed installed hardware, i.e., conveyors, an ASRS, and industrial robots, so that it is not easily scalable. Adapting the throughput capacity always requires an extensive modification of the infrastructure, which is costly and not realizable on short notice. On the positive side, automation allows to handle even large orders from medium-sized assortments with the necessary efficiency and it is one of the few systems which can realize store-specific build-ups (provided that they do not collide with stability issues). Stability issues and store-specific build-ups can be considered when generating the packing pattern for the pallets and roll cages and the cases merely have to be retrieved in the respective sequence. Most real-world implementations of fully-automated case picking are reported for large food store chains (Azadeh et al., 2019).

Layout design: when setting up a new case picking system or restructuring an existing one, selecting the right equipment and dimensioning it appropriately is among the most important challenges. Suited choices for all types of equipment, i.e., depalletizers, the ASRS, and palletizing robots, have to be made in a concerted manner, such that bottlenecks obstructing the targeted throughput are avoided. Note that this is a general problem in all warehousing systems, which is discussed in more detail in Section 9. With regard to the ASRS technology, the main alternatives are traditional crane-operated (high-bay) racks (Boysen & Stephan, 2016; Roodbergen & Vis, 2009) or carousel systems (Litvak & Vlasiov, 2010). A comparatively novel competitor are lift-and-shuttle ASRS (see steps 2 and 3 of Fig. 3). They separate the storage and retrieval of goods among shuttles and lifts for horizontal and vertical movement, respectively (Carlo & Vis, 2012). This division of



Fig. 3. Fully-automated case picking.

labor increases the throughput and reduces the investment costs, because for the movable devices rather standard one-dimensional equipment can be utilized. Thus in the recent years, there is a trend toward these systems. Existing research on ASRS is reviewed by the survey papers of Roodbergen and Vis (2009) and Boysen and Stephan (2016). Most existing research, however, focuses on ASRS exclusively (Roodbergen & Vis, 2009), and does not integrate connecting material handling devices and interdependent stages of the total order fulfillment process. In the case of fully-automated case picking systems, however, this omission of existing research is especially dangerous, because all components together determine the throughput capacity of order fulfillment.

Storage assignment: the cases are stored inside the ASRS, so that each incoming case has to be assigned a storage position inside the racks of the ASRS. The alternative storage policies typically applied in warehousing in general and ASRS in particular are surveyed in (De Koster et al., 2007) and (Roodbergen & Vis, 2009), respectively. The general aim of these storage policies (e.g., class-based storage) is to assign frequently demanded goods storage positions closer to the input/output point, so that quick storage and retrieval moves are enabled and thus higher throughput capacity of the ASRS is gained. Fully-automated case picking, however, constitutes a special case. Here, cases – once retrieved – do not return back into the ASRS. They are the smallest unit size and are completely absorbed by their dedicated orders. Thus, incoming cases not expected to be demanded in the near future should be stored farther away from the input/output point, so that preferred storage space is not blocked by slow movers. However, the storage assignment of each case is not an isolated decision, because cases of the same SKU can substitute each other. The peculiarities of fully-automated case picking during storage assignment have not been considered in the existing literature yet. The only exception is by Chen, Langevin, and Riopel (2010). They consider the assignment of goods in a crane-operated high-bay rack operated in dual commands (i.e., the crane combines storing a product and returning a demanded piece in between any pair of subsequent visits at the input/output point, e.g., see (Boysen & Stephan, 2016)). For this setting, Chen et al. (2010) assign storage positions to products whose storage intervals are not bound to uncertainty, but known and given. Stochastic approaches and suited storage policies for uncertain demands are yet missing.

Order processing: once a specific order of a brick-and-mortar store is selected for processing, the outstanding planning component, which determines all other operational processes, is the packing pattern derived by an optimization procedure. Given the demands of the store, the requested cases have to be partitioned among shipping carriers. Note that the most widespread shipping carriers are pallets and roll cages. Since most practical applications of fully-automated case picking are reported for large food retailers (Azadeh et al., 2019) who predominantly apply roll cages, we will apply this term here as a placeholder for all possible shipping carriers. Once the cases per roll cage are determined, the detailed packing pattern has to decide on the exact placement of each case in relation to all other cases placed on the same roll cage. Given the detailed packing pattern, it can be translated into a retrieval sequence of cases, because, naturally, cases to be placed bottom-most by the palletizing robot have to arrive first. Thus, the detailed packing pattern and the resulting retrieval sequence of cases determine the sequence in which the pieces have to be retrieved from the ASRS. Consequently, there is not much flexibility for sequencing storage and retrieval moves within the ASRS (Boysen & Stephan, 2016; Litvak & Vlasiov, 2010), so that we concentrate our survey on the packing aspect.

Packing optimization has a long-lasting tradition and is still a vividly researched field. Survey papers on the wide range of alternative problem settings are provided by (Bortfeldt & Wäscher, 2013; Dyckhoff, 1990; Ram, 1992; Wäscher, Haußner, & Schumann, 2007). The packing problem specifically relevant in the fully-automated case picking context has the following characteristics:

- **3D:** first, we have a three-dimensional packing problem with heterogeneous cases. This excludes the wide-range of two-dimensional packing problems (Dowsland & Dowsland, 1992; Lodi, Martello, & Monaci, 2002a), because with differently sized cases the packing process can hardly be decomposed into layer-wise (two-dimensional) packing problems. The latter is only possible, if the assortment of cases is standardized and rather homogeneous.
- **Objective:** to reduce transportation costs, the typical objective of the packing process is to load the given demand of each store onto a minimum number of roll cages. Note that other



(a) A pick-and-pass system for small SKUs, here: GEBHARDT Pick To Light system



(b) A trolley-line picking system, here: CaddyPick™ by Swisslog

Fig. 4. Pick-and-pass systems for small (left) and large (right) SKUs.

objectives, i.e., a store-specific build-up or stable loads, are rather translated into (soft or hard) constraints. This adds the bin packing characteristic to our packing problem, i.e., we have identical load carriers and want to utilize a minimum number of them for packing all given cases. Important methodological contributions for solving the three-dimensional bin packing problem are, e.g., from (Faroe, Pisinger, & Zachariassen, 2003; Kang, Moon, & Wang, 2012; Lodi, Martello, & Vigo, 2002b; Martello, Pisinger, & Vigo, 2000). In addition to height restrictions up to which each roll cage can be packed, there also exist weight restrictions to avoid too heavy roll cages (Bortfeldt & Wäscher, 2013).

- **Temperature classes:** food retailers also have to consider different temperature classes. If each case has a specific temperature class, then the overall problem decomposes into one packing problem per class. However, if some cases may be transported in alternative temperature classes, then an even more complicated packing problem including substitution is to be solved.
- **Orientation:** in the warehousing context, most products are non-rotatable. This means that for safety and stability reasons cases need to be placed in a predetermined way, e.g., onto the bottom of a beverage crate. Crainic, Perboli, and Tadei (2009), for instance, introduce a two-level tabu search algorithm for the 3D-BPP with non-rotatable products.
- **Stability:** when packing cases onto roll cages, it is furthermore necessary to avoid crushing damage which threatens whenever heavy cases (e.g., beverage crates) are placed on fragile ones (e.g., egg cartons). Many cases, thus, have a maximum bearing strength, which defines the maximum pressure that can be applied over the top face of a case (Bortfeldt & Wäscher, 2013). Optimization procedures integrating this aspect are, e.g., provided by Bischoff (2006) as well as Junqueira, Morabito, and Yamashita (2012).
- **Build-up:** finally, we have the store-specific build-up, which (if applied by the specific retail chain) suggests to derive the packing pattern from the layout of each store (as long as this does not collide with stability aspects, see above). This leaves not much flexibility when determining the sequence of cases arriving at the palletizer, so that the solution space for an optimization problem is considerably reduced, if this characteristic is relevant.

While there exist research papers for most of these characteristics in isolation (or even combinations of two or three aspects), we are not aware of papers dealing with all of them simultaneously. In practice, multiple proprietary solution have been developed. The

authors are aware of a system based on constructive heuristics improved by local search. Thus, there is further room for future research in this area.

4. Pick-and-pass systems for large orders

The basic hardware element of a pick-and-pass system is a conveyor that transports, accompanied by a human picker, the order carriers (e.g., bins) in which customer orders are collected along the shelves of a fixed picking path. Different kinds of pick-and-pass systems exist, depending on the size of SKUs to be handled. The typical workstation for small-sized SKUs, such as pharmaceuticals, is depicted in Fig. 4(a). Here, the conveyor system is a simple unpowered roller-belt conveyor where the picker has to push the bin or carton associated with the current picking order along the picking path. To transport the order carriers between these workstations the belt is normally powered. Often, the shelves of a pick-and-pass workstation are equipped with a pick-by-light mechanism indicating the requested pieces per SKU of the current order. Obviously, these systems struggle if large-sized SKUs and/or large store orders with many pieces per order line have to be handled. For these cases, a special pick-and-pass system, the trolley-line picking system, has been developed. This system is able to transport up to two pallets or roll cages per order. The trolleys are automatically transported along the picking path while hanging from a monorail conveyor mounted under the ceiling. Such a system is, for instance, merchandised by Swisslog where it is called the CarryPick™ system, see Fig. 4(b).

Note that another alternative for erecting pick-and-pass systems for large orders is the application of traditional automated guided vehicles (AGVs) driving on the ground along a fixed path of shelves. Such a system is reported by Füßler, Fedtke, and Boysen (2019a) for a warehouse handling beverage crates. Although (strictly speaking) it applies no trolleys, we will refer to it as a trolley-line picking system too. These systems for larger orders are better suited in the context of brick-and-mortar retailing, so that we will concentrate the following descriptions of the system setup on trolley-line picking systems. The trolleys, while being transported along the picking path, are accompanied by human logistics workers, who put the requested number of pieces per SKU onto a trolley. Once a trolley reaches a requested SKU when moving along the given path, it automatically stops next to the SKU and announces the requested number of pieces on a display. Often, requested SKUs are also indicated by a light signal, so that a picker can conveniently locate a pick request. A picker moves toward the respective position, puts the requested pieces onto the trolley and

accredits the pick, e.g., by pushing a button. Trolleys can also be equipped with an automatic weighing mechanism to reduce picking errors (Swisslog, 2017). Then, the trolley moves onwards toward the next SKU and so on, until the associated store order is completed. Finally, the trolley moves automatically onwards to the shipping area, while the picker returns in backward direction to the subsequent trolley.

The main difference between trolley-line picking and a pick-and-pass system based on a roller conveyor is that the roller conveyor is unpowered (in a workstation), so that the picker has to accompany the order bin (or carton) along the picking path. Trolleys, instead, move independent of the picker, which opens further possibilities to increase picking performance. A main operational problem of both settings, however, is the blocking aspect. Trolleys utilizing the same monorail, AGVs traveling along the predefined picking path in a narrow aisle, and bins being pushed along a roller conveyor cannot overtake each other, so that order carriers waiting for service block subsequent orders. The blocking aspects also delimit pick-and-pass systems from other fixed-path picking systems, such as milk-run picking and polling systems. In these systems, pickers repeated follow a fixed path. At each location passed, the pickers are displayed the total number of pieces demanded by the orders that have arrived since their previous visit (Van der Gaast, De Koster, & Adan, 2019; Gong & De Koster, 2008). However, these systems require an additional consolidation stage to sort the orders. They are rather suited for small-sized order structures of on-line retailers (Boysen et al., 2019b). In pick-and-pass systems, each order carrier is dedicated to a specific store order, so that we have a sort-while-pick system (De Koster et al., 2007) and do not require an extra consolidation stage. A survey on these alternative fixed-path systems is provided by Boysen et al. (2019b).

Trolley-line picking fulfills all our seven requirements except for one. It struggles with store-specific build-ups. The trolleys move along a predefined path, so that the SKUs are passed in a given sequence once their storage assignment is fixed. Stores with different layouts require different build-ups and, thus, different sequences of SKUs, which cannot be altered on short notice for each new order. Stability aspects, however, are not store-specific and can easily be ensured by ordering the SKUs according to decreasing weight. Although the system applies some fixed installed hardware, e.g., the conveyor system, it leaves some flexibility to adapt throughput capacity. By adding more bins or trolleys and pickers, throughput capacity can be extended on short notice to process peak demands. Applications of trolley-line picking are reported for different food store and drugstore chains (Swisslog, 2016). Füßler et al. (2019a) treat the case of German distribution center that supplies liquor stores, supermarkets, and large restaurants with beverages. They report on an AGV-based trolley-line system. Traditional pick-and-pass systems additionally struggle with large-sized orders. Therefore, their application is limited to small-sized SKUs such as pharmaceuticals or cosmetics.

Layout design: foremost, the following long-term decisions have to be taken during layout design planning: the sizing of the picking path, the trolley fleet size (if applied), and the division of labor among multiple pickers. Note that the speed of trolleys is not among the critical levers during layout design. Trolleys share the shop floor with human pickers, so that, for safety reasons, the maximum trolley speed is standardized to walking speed. Further note that for pick-and-pass systems with sequential zoning, dedicated to small-sized SKUs, the most important layout decisions (e.g., number and sizes of pick-and-pass workstations and where to have bypasses) and their interdependencies are treated by Van der Gaast, De Koster, Adan, and Resing (2020).

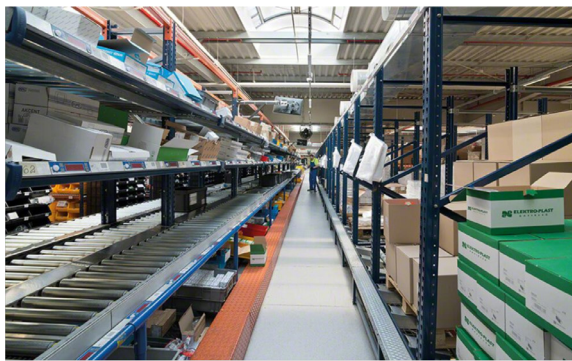
- **Storage capacity:** the length of the picking path determines the number of SKUs that can be stored in the system. The capacity

should rather be dimensioned a bit larger, because this leaves some space if the assortment of SKUs expands over time. Unused capacity at the end of the line is unproblematic and only prolongs the unattended travel of the trolleys toward to shipping area. In traditional pick-and-pass systems this finding does not hold, because a prolonged picking path also enlarges the distance the picker has to accompany each order, which may deteriorate picking performance. This trade-off is analyzed by Van der Gaast, De Koster, and Adan (2018).

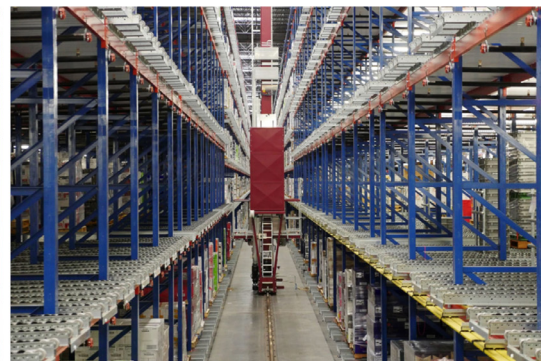
- **Trolley fleet:** the throughput capacity cannot be increased just by increasing the number of trolleys. Since trolleys have to be filled by human pickers, the trolley fleet and the picker workforce have to be determined in an interdependent manner.
- **Division of labor:** to share labor among multiple pickers there are three general alternatives. Either each picker traverses the complete picking path, so that parallel pickers work in a shifted manner, or the line is separated into disjunct segments each operated by a dedicated picker. The latter is also denoted as sequential zoning. Under the former policy, pickers have an excessively long way back along the complete line. Furthermore, there is the additional alternative of a dynamic workload sharing, e.g., by applying the bucket-brigade protocol (see (Armbruster, Gel, & Murakami, 2007; Bartholdi & Eisenstein, 1996; Bartholdi, Eisenstein, & Foley, 2001)). Under this policy, pickers having completed an order return against the flow direction until they meet with a predecessor and take over their current order. In this manner, the takeover carries forward along the line, until, at the start, the first picker starts with a new order. A performance comparison of bucket brigades and sequential zoning in the warehousing context is provided by Koo (2009). Separating a pick-and-pass system for small-sized items into fixed zones has been treated by Jane (2000) and Jane and Lai (2005). They propose heuristic algorithms to smooth the workload imbalance among a given number of zones. Otto, Boysen, Scholl, and Walter (2017) integrate ergonomic aspects into this decision, such that the ergonomic strain is fairly distributed among pickers. Jewkes, Lee, and Vickson (2004) consider sequential zoning for a related order picking system where each picker is assigned a specific home base to which order bins are conveyed by a powered conveyor belt. A pick-and-pass system with home bases is also investigated by Yu and De Koster (2008) and Melacini, Perotti, and Tumino (2011). Both papers apply queuing models for performance analysis of different pick-and-pass layouts, e.g., with different zone numbers. All these previous papers address pick-and-pass systems for small SKUs; the peculiarities of trolley-line picking have (to the best of the authors' knowledge) not been treated yet.

Storage assignment: in the pick-and-pass context, the storage assignment decides on the arrangement of SKUs along the picking path. In a system based on multiple zones the storage assignment, thus, decides in which zone a specific SKU is to be picked. This may produce work imbalance and bottlenecks, if some zones receive all fast-moving items and others only the slow movers. As elaborated above storage assignment has to be jointly treated with zoning, which is done by (Jane, 2000; Jane & Lai, 2005; Jewkes et al., 2004; Otto et al., 2017; Yu & De Koster, 2008). Furthermore, Pan and Wu (2009) provide analytic models for performance evaluation, and Pan, Shih, Wu, and Lin (2015) present a genetic algorithm for the storage assignment decision in a multi-zone setting.

In trolley-line picking systems where orders are moved automatically, even the storage assignment of a single zone is an optimization problem. In the worst case, at least one of an order's requested SKUs is stored at the beginning and another SKU at the end of the picking path. In this case, a picker has to accompany the respective trolley along the complete line, so that nothing is



Step 1) Ground level order picking along a roller belt conveyor in front of the rack by a human picker in a system of Mecalux



Step 2) Swapping of unit loads behind the racks by a crane of Westfalia Technologies

Fig. 5. Crane-supplied pick face.

won compared to traditional pick-and-pass system. Reduced picker walking is only possible when all SKUs of an order are stored closer together. This allows the picker to bring forward the return to the next order, so that a more efficient picking process is enabled. Füßler, Stephan, and Boysen (2019b) treat this optimization problem for a deterministic order set. They define the resulting storage assignment problem, prove complexity, and also consider the duplication of frequently requested SKUs along the path. The latter increases the flexibility and opens a choice among storage positions from which a specific SKU is picked. The authors show that duplicating the right SKUs and placing them wisely allows to greatly reduce the picking path of a single picker. The work sharing among multiple pickers or storage assignment policies if only stochastic information on the SKU demands is available have, however, not been considered yet. For a related milk-run system, Van der Gaast et al. (2019) suggest a genetic algorithm for placing the SKUs with given order probabilities along a fixed picking path.

Order processing: in traditional pick-and-pass systems with unpowered roller conveyors pickers have to accompany the order along the complete picking path of their zone anyway. Thus, there are no operational levers to improve picking performance. In a trolley-line picking system this is different, so that our following descriptions focus on this setup. Given a set of customer orders to be processed next, determining the order sequence is the most important operational decision task. In general there are two alternative picking policies under which a trolley-line picking system can operate in each single picking zone (Füßler et al., 2019a):

- When applying the *order-by-order policy* a picker has to accompany the complete order from first to last picking position of the current order before switching to the next one. Under this policy, picking performance can be increased by an order sequence, which ensures that the way back from the last picking position of the predecessor order to the first picking position of the successor order is reduced. This problem can be solved to optimality by applying the famous algorithm of Gilmore and Gomory (1964), which solves a special case of the traveling salesman problem on the line in polynomial time. Note that the latter is only true, if minimizing the makespan is addressed. In the warehousing context, however, lateness-oriented objectives also play an important role, so that there is room for future research.
- The *order-swapping policy* allows the picker to flexibly swap among partial orders within her zone, so that first all subsequent trolleys queuing in a specific segment of the path can be serviced before walking to another segment. In this case, order

sequencing is interdependent with an additional picker routing problem between the picking positions of waiting trolleys blocking each other along the path. This problem has been considered by Füßler et al. (2019a). They decompose the problem, determine picker routes for given order sequences, and apply these exact and heuristic algorithms in a larger heuristic framework by also varying the order sequences. They compare the results with optimal order sequences for the order-by-order policy and conclude that order-swapping allows to greatly increase the picking performance of a single picker.

Future research should compare both order processing policies for typical trolley-line picking systems that deploy more than a single picker per zone and how the decisions interact among successive zones. A systematic comparison on the performance gains if the orders travel automatically (trolley-line picking) or need to be moved by the picker (pick-and-pass systems for small-sized SKUs), is yet missing too.

5. Crane-supplied pick face

The basic element of a warehouse applying a crane-supplied pick face is a rack, which is subdivided into two parts: the bottom part is used as pick face and the upper as reserve area. The bottom row is the picking level where SKUs are directly accessible by human order pickers. Parallel to the pick face is a conveyor, e.g., an non-powered roller conveyor, on which bins or cardboard boxes associated with a store order accompany a human order picker along the pick face (step 1 of Fig. 5). The picker and the current bin move along the pick face, which may be separated into multiple zones each operated by another picker, and collect all SKUs demanded by the current picking order. Once the end of the pick face is reached and the current picking order is completed, the conveyor transports the order to the shipping area and the picker returns to the start where a new picking order is initiated (or taken over from the previous zone). Note that often a pick-to-light system is applied to ease the orientation of pickers. Such a system indicates all SKUs requested by the current order with a light signal. Alternatively, also crane-supplied pick faces exist where the order bin remains stationary and the picker has to fetch the demanded SKUs along the pick face. However, if many SKUs are demanded by brick-and-mortar stores which cannot all be carried by the picker on a single tour, such a system causes excessive unproductive picker walking and is, thus, preferred in the e-commerce context where orders are small.

Behind the racks operates a crane that automatically exchanges SKUs between the bottommost pick face and the reserve area of

upper levels (step 2 of Fig. 5). To reduce the walking distance of the picker, the pick face does not contain the complete assortment of SKUs. Only a subset of SKUs is *actively* loaded into the pick face at a time. If a current picking order requires a SKU, which is only available in the reserve area we call this SKU *passive*. In this case, the crane has to remove an active SKU currently not required from the pick face and switch it with the respective (formerly passive) SKU.

Crane-supplied pick faces, also known as ASRS with multiple-in-aisle pick positions (Ramtin & Pazour, 2015) and dynamic storage systems (Yu & De Koster, 2010), seem especially suited if each store order contains not too many order lines and the assortment of SKUs is not overly large as well. The former allows to reduce the length of the pick face, such that picker travel is reduced, and the latter reduces the capacity of the reserve area where excessive demand for space considerably increases investment costs. Note that, naturally, the exact threshold where order sizes become too large to be efficiently handled by a crane-supplied pick face is hard to determine. In their empirical study among 68 Italian warehouses Dallari, Marchet, and Melacini (2009), for instance, locate this threshold for pick-to-box systems, to which crane-supplied pick faces have to be counted, at an average order volume of 0.5m³. More research is certainly required in this direction. To adapt throughput capacity on short notice, pickers can flexibly be added to or removed from the system. If multiple picker work in parallel, either each picker always traverses the complete pick face, so that they move in a shifted manner, or the lane is subdivided into disjunct areas each operated by a different picker. In both cases, however, adding additional pickers is only sensible up to a certain level. Once the crane is the bottleneck resource, additional pickers cannot increase performance. Since the crane's performance, e.g., by switching to a faster crane generation, or by additional cranes sharing the workload behind the racks, cannot be changed on short notice, crane-supplied pick faces struggle with varying workloads. Moreover, stability aspects and store-specific build-ups are barely realizable. In addition to the switches between passive and active SKUs, this would add the need to switch the storage positions of active SKUs along the pick face. This, however, increases the waiting time of the picker for the crane and degrades picking performance. Applications of crane-supplied pick faces are reported for the cosmetics industry (Kim, Heragu, Graves, & Onge, 2003), deep-freeze warehouses of food retailers (Ramtin & Pazour, 2014), and a book distributor applying them for fast-moving products ordered by brick-and-mortar book stores (Schwerdfeger & Boysen, 2017).

Layout design: The main decision of when setting up a crane-supplied pick face is its sizing, which faces the following elementary trade-off: A larger pick face has additional capacity to store active SKUs. This relieves the crane, but causes additional walking for the picker who has to operate a larger pick face. On the other hand, a smaller pick face reduces picker walking, but increases the workload of the crane for switching SKUs. Too many switches may cause idle time for the picker and the picking performance decreases. To the best of the authors' knowledge this trade-off has not been treated in the context of crane-supplied pick faces. Existing research rather treats extended layout decisions. Ramtin and Pazour (2014) develop different analytical methods to quickly evaluate different layout options of large multi-aisle systems applying many pickers in parallel. Specifically, they evaluate erecting an additional elevated picking floor on a mezzanine that enables two-level order picking. Furthermore, they evaluate a postponement of newly arriving unit-loads dedicated to the replenishment area to off-peak hours.

Storage assignment: the storage positions of SKUs along the pick face are not kept stable over a longer period of time. Since the capacity of the pick face is not large enough to store all SKUs, they

have to be switched continuously by the crane. Due to these continuous changes, a synchronization with the storage positions of SKUs in the reserve area is also barely possible. Therefore, storage assignment is rather an operational problem. Analytical models estimating the resulting crane travel for different storage positions of SKUs along the pick face are derived by Ramtin and Pazour (2015). Given their analytical model, the authors evaluate different storage assignment policies for real-world warehouses.

Order processing: behind the rack, crane scheduling for a given set of switches in a crane-supplied pick face is treated by Kim et al. (2003) and Boysen and Stephan (2016). The former paper deals with a huge CSPF system in the cosmetics industry and focuses on efficient crane scheduling. Different heuristic procedures are provided to reduce the makespan of the crane when switching bins. Crane scheduling is also addressed in (Boysen & Stephan, 2016). Here, a complexity proof is provided that proves NP-hardness for the special case of the traveling salesman problem where all crane moves start or end at the bottommost pick face.

A holistic problem setting combining the front and back view is addressed by Schwerdfeger and Boysen (2017). They treat order sequencing and determining the switching plan of SKUs that have to be exchanged by the crane with any new order in a holistic problem setting. If plenty SKUs have to be switched between two successive orders, this may cause waiting time of the picker for the crane to process these switches. Which tools to switch also depends on the order sequence, so that both problems jointly impact picking performance. Schwerdfeger and Boysen (2017) give a formal problem description of the SKU switching problem and prove NP-hardness with a reduction from the Hamiltonian Path problem. They introduce a heuristic decomposition procedure and a branch and bound algorithm with various lower bounds and investigate the computational performance of their algorithms. They also stress the close relationship of the SKU switching problem with the job sequencing and tool switching problem of flexible manufacturing (for a survey on this problem see (Calmels, 2019)). Here, the tool magazine of a machine needs to be automatically equipped with tools required by jobs to be successively processed. The existing solution approaches of this field, however, only identify which tools (SKUs) are to be switched, but not their exact positions in the magazine (along the pick face). Since the storage position of the SKU can influence the waiting time of the picker, additional research is required in this direction. Stochastic analysis is required if the order set is not known ahead, but orders arrive randomly over time. Analytical equations if order arrivals follow a Poisson distribution are developed by Yu and De Koster (2010). Their analytical model can be applied to evaluate differently sized pick faces and the inherent trade-off between picker travel and crane effort. They show that a properly dimensioned system can considerably increase throughput and reduce labor effort at the same time.

Especially, the aforementioned trade-off between the number of SKU switches and a picker's walking distance depending on the size of pick face deserves being investigated by future research.

6. Bulk picking and store consolidation

Bulk picking separates the order fulfillment process into three basic stages, retrieving SKUs from storage, inducting pieces onto a sorter, and order consolidation. When processing a wave of orders, i.e., a subset of the total order set jointly sorted in one consolidation run, unit loads of SKUs (typically bins) are successively retrieved either from a parts-to-picker system (ASRS, e.g., a crane-operated rack (Boysen et al., 2018a; Roodbergen & Vis, 2009), a carousel (Litvak & Vlasiov, 2010), or a lift-and-shuttle system (Carlo & Vis, 2012); step 1a of Fig. 6) or an interconnected parts-to-picker system, where human order pickers receive the demanded



Fig. 6. Bulk picking.

pieces from shelves (step 1b of Fig. 6). In both cases, bins of SKUs are delivered to one or multiple parallel induction stations (step 2 of Fig. 6). Here, a human logistics worker is displayed the total number of pieces of the current SKU demanded by all orders of the active wave, removes them from the unit load, and places one piece after the other on the consolidation conveyor. Note that if the SKUs arrive from a parts-to-picker system, then the human order picker typically has already retrieved the requested number of pieces, so that the complete bin is inducted. Further note that for some warehouses also the application of fully-automated induction technology is reported (Gallien & Weber, 2010). Once all pieces of the current SKU are retrieved and placed onto the conveyor, the unit load is returned into the storage system or the empty bin is removed. In this manner, all SKUs demanded by the active wave are successively processed at the second stage by the logistics worker.

The main hardware element of the third stage (step 3 of Fig. 6) is a consolidation sorter, which consists of a bomb-bay, sliding-shoe, crossbelt, or tilt-tray sortation conveyor (see Boysen, Briskorn, Fedtke, & Schmickerath, 2019a) leading into packing stations. After passing a recognition station, where pieces are identified, e.g., by scanning a bar code, the belt moves along a sequence of successive packing stations. Each packing station is immediately assigned to a store and collects the demanded pieces of the respective store order. Whenever a piece reaches its dedicated packing station it is redirected and collected there. Once an order is complete, a light signal indicates this status, so that a packing worker can approach the station, pack the pieces into a shipping carton or roll cage, and forward it to the shipping area. Then, the packing station is released and can be assigned to another store order of the successive wave. In this way, wave after wave is processed until all store orders are fulfilled.

Especially the consolidation conveyor is hardly scalable on short notice, so that bulk picking struggles with varying workloads. Moreover, store-specific build-ups are barely realizable. Since SKUs are retrieved in a batchwise manner; no individual SKU sequences for each store can be processed. To realize store-specific build-ups in a bulk picking environment, either the packing workers at the packing stations have to assemble them in a laborious manual process or another sorting system is required. A comparatively novel technology are bag or pouch sorting systems (see Fig. 7), which carry individual small-sized pieces in small bags hanging from trolley conveyors. These systems allow to re-sequence the bags in a system of switches, so that individual SKU sequences can be realized in between both stages. Stability aspects, however, are easily realizable also with conventional loop conveyors. The SKUs merely need to be diverted in the respective order, e.g., according to decreasing weight and stability. All other requirements for brick-



Fig. 7. Bag or pouch sorter system (Source: Dürkopp).

and-mortar stores are fulfilled. Applications of bulk picking are reported for retail chains for shoes, video games, and office supplies (Vanderlande, 2018).

The biggest threat when applying a bulk picking order fulfillment arises when, in order to avoid idle time at any of the stages, subsequent waves are released in too quick succession. In this case, it may occur that all packing stations are still occupied by not yet completed orders of the previous wave and further pieces arrive for new orders of the next wave that cannot be assigned a packing station. This produces *aimless pieces* without a target packing station. On the other hand, postponing wave succession leads to idle in the order picking stage (stage 1b in Fig. 6 and the induction stations (step 2 in Fig. 6). When waiting even longer also the packers may become idle, because at the start of any new wave it typically takes some time before the first store orders have collected all their demanded SKUs. Aimless pieces cause double-handling and plenty manual overhead, where the extent mainly depends on the kind of consolidation conveyor applied (Boysen et al., 2019a):

- If a *linear* consolidation conveyor is applied (for an example see Boysen, Fedtke, & Weidinger, 2018b), aimless pieces have to be collected at the end of the line. They either need to be manually sorted or reinserted into the consolidation process. In addition to the manual double-handling both reactions increase the risk of picking errors.
- An alternative is a *closed-loop* consolidation conveyor (for examples see Johnson & Lofgren, 1994; Johnson, 1998). Here, aimless pieces can recirculate in the loop until some store order is completed and another packing station for a subsequent order is available again. However, if more and more orders have to recirculate until the complete loop is fully occupied, no further pieces can enter the conveyor to complete not yet

finished orders, no packing stations can be freed, and a gridlock occurs (see also, Gallien & Weber, 2010). The recovery after such a gridlock can be very laborious and is error-prone.

Thus, avoiding aimless pieces due to a completely occupied packing area while not causing excessive idle time in previous stages is among the top priorities when solving the following main decision problems of bulk picking.

Layout design: when setting up a bulk picking system, sizing the consolidation conveyor and selecting a suited kind of conveyor technology are the outstanding decisions (Boysen et al., 2019b). The former decision includes the length and velocity of the conveyor as well as the number of packing stations. A detailed survey on the application of consolidation conveyors is provided by Boysen et al. (2019b). Johnson and Lofgren (1994) study the conveyor belt speed and the sizing of a loop conveyor by modeling the design and processes of Hewlett-Packard's new North American distribution center. One of their results is that overlong conveyor loops (i.e., to protect against gridlocks) come at the price of deserted packing stations. Russell and Meller (2003) compare automated loop conveyors with manual sorting processes. They introduce a cost based optimization model for comparing the total annual costs and apply their model to different warehouse settings. In their sensitivity analysis, it is shown that the recommended system configuration does not solely depend on a single parameter, but requires a complex decision hardly generalizable. Bozer and Sharp (1989) compare different conveyor layouts and evaluate the throughput performance of closed-loop and linear consolidation conveyors when applying different numbers of packing stations. Future research should also consider bag sorters and benchmark this novel technology with conventional sorting technology.

Storage assignment: when applying a bulk picking system, storage assignment is less of an issue. If a parts-to-picker system is applied on first stage (step 1a in Fig. 6), it has to be ensured that the ASRS delivering the SKUs to the induction stations is not the bottleneck of the process. A suited storage assignment of frequently requested SKUs closer to the input/output point within the ASRS can help to avoid this. Applying an ASRS in a bulk picking warehouse, however, adds no peculiarities, so that we refer to existing survey papers on ASRS and suited storage assignment policies instead (see (Boysen & Stephan, 2016; Roodbergen & Vis, 2009)). Analogously, applying a picker-to-parts system on first stage (step 1b in Fig. 6) does not add much to the existing storage assignment policies of these systems, so that we refer to the relevant survey papers (Cormier & Gunn, 1992; De Koster et al., 2007; Gu et al., 2007; 2010).

Order processing: at an operational level, the main lever for the delicate steering problem having to trade off idle time against the threat of gridlocks, is the succession of wave releases elaborated above. In existing research, this problem has mainly been treated in a slightly different setting rather relevant for online retailers. Their warehouses often also apply a picker-to-parts system on stage 1 (step 1b in Fig. 6), but do not pick all demanded pieces of a specific SKU into homogeneous bins, but apply a batching and zoning policy. The bins each containing partial orders dedicated to multiple customers are assembled by human pickers in multiple zones in parallel and arrive in the consolidation area where they are sorted with the help of consolidation conveyors (Petersen, 2000). Order consolidation in an e-commerce warehouses is reviewed by the survey papers of Boysen et al. (2019b) and Boysen et al. (2019a). Thus, the main distinction is that in bulk picking all pieces of the same SKU arrive in direct succession, whereas in the e-commerce context widely mixed sequences of SKUs are the norm. In the e-commerce context, Bozer, Quiroz, and Sharp (1988) and Johnson and Lofgren (1994), for instance, suggest that

the next wave should be released whenever 90% and 50% of the previous wave have been completed in the consolidation area, respectively. Gallien and Weber (2010) investigate a waveless order release from the first stage with queueing models and investigate whether such an approach is superior to a wave release. Kizilaslan, Bayraktar, and Eldemir (2013) consider the sizing of waves with the help of analytical models. Boysen et al. (2018b) also consider the sequence, in which the bins of a wave are released from an intermediate ASRS toward the consolidation area and conclude that a suited sequence can reduce the risk of a congested consolidation stage. Whether, however, the existing results, all obtained for a zoning and batching environment of a picker-to-parts warehouse on stage one, are in fact transferable to a bulk picking environment lacks scientific support yet. In a bulk picking system, demanded pieces are released in a SKU-wise manner down the line, whereas in the former setting widely mixed piece sequences are sent down the conveyor. How this impacts the performance of order consolidation should be investigated by future research.

Another operational decision problem occurring during order processing in a bulk picking environment is the assignment of packers to packing stations of the consolidation conveyor (Boysen et al., 2018b). They can either each be assigned a fixed area of dedicated packing stations. The alternative is a dynamic assignment where packers freely move from station to station whenever they are available and a beacon indicates a completed order not yet processed. The latter leaves more flexibility and should thus increase consolidation performance, but comes at the price of additional packer walking. Moreover, assigning errors where pieces are placed into the wrong shipments to culprit workers is harder and requires more organizational overhead. An in-depth comparison of both packer assignment policies in a bulk picking environment is yet missing.

7. Put systems

Put systems (also denoted as inverse order picking, put-to-light systems, or order distribution systems) invert the basic logic of picker-to-parts systems in general and pick-and-pass systems in particular. Traditionally, the SKUs are stored at fixed storage positions and human pickers successively visit selected positions while collecting the products defined on a pick list. In a put system, the SKUs are mobile while the carriers in which customer orders are collected remain stationary. Note that for a better distinction we distinguish *SKU bins* and *order bins*. The former contain homogeneous pieces of a specific SKU and are moved along the put lane by a human picker. The latter are associated with a picking order of a store and collect the picked pieces at a fixed position. Note that most of the systems we know apply bins for collecting orders, so that we will speak of order bins, well knowing that systems based on other order carriers, e.g., roll cages exist.

At the start of the put lane, there is a conveyor connecting an ASRS (e.g., a crane-operated rack (Boysen & Stephan, 2016; Roodbergen & Vis, 2009), a carousel (Litvak & Vasiou, 2010), or a lift-and-shuttle system (Carlo & Vis, 2012)) or a manual picker-to-parts picking system, where the demanded number of pieces per SKU has been retrieved from a unit load and put into a SKU bin. The conveyor delivers the SKU bins on demand toward the put system. Here, a human order picker takes over the bin and moves it, typically on a non-powered roller conveyor, along the order bins positioned in racks. The SKU bin is identified, e.g., by scanning a bar code, and put-to-light displays (step 1 of Fig. 8), mounted directly next to the respective order bins, indicate the demands. That is why these systems are called put systems (Ten, Hompel, & Schmidt, 2007). A human picker satisfies an indicated demand, puts the requested number of pieces into the order bin, and accredits the current SKU by pressing a button on the put-to-light



Step 1) Announcement of demand, e.g., by a pick quantity display of invata



Step 2) Inverse order picking, e.g., in a put-to-light system of Lightning Pick

Fig. 8. Inverse order picking.

display. In this way, SKU bin after SKU bin is moved along the lane. Completed order bins are forwarded to the shipping area, e.g., by pushing them on another conveyor running behind the rack. A freed storage position can be replaced by a new order bin associated with the same or another (not yet processed) store. There are different ways how to divide labor in put systems:

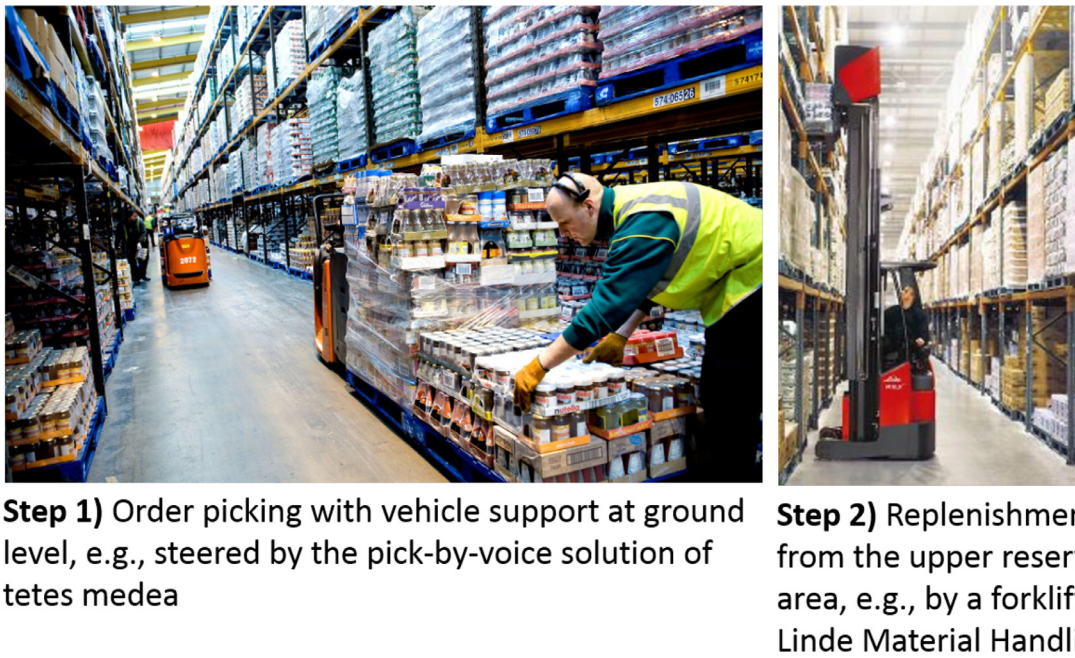
- In the most basic case, a single picker operates the complete system alone. The picker always accompanies each SKU bin along the complete lane. When reaching the end of the lane, she moves back to the start and proceeds with the next SKU bin.
- To increase throughput capacity multiple pickers can be applied in parallel in a shifted manner. Each picker still traverses the complete lane only separated by a few meters. A major operational problem may be a blocking among pickers if their working speed varies.
- Alternatively, the lane can be subdivided into different areas each operated by a single picker (see step 2 of Fig. 8). Areas can be disjunct, so that each picker has a fixed range of order bins in her area, or flexibly depend on the current workload. The latter can, for instance, be achieved by the famous bucket brigade protocol (Bartholdi & Eisenstein, 1996), see Section 4.

The possibility to add multiple pickers is the reason why put systems can flexibly be adapted to varying workloads. Since the racks and roller conveyors are rather low tech, the system is also quickly adaptable if the number of stores changes on a mid-term horizon. However, put systems struggle with very large store orders, especially if the SKUs are heavy and bulky. The human order pickers still have to be able to move the SKU bins along the roller conveyor and if some orders require an extraordinary number of pieces, the capacity of the SKU bins may not be large enough to carry enough of them. Store-specific build-ups of the shipments are not realizable, because all stores receive the SKUs in the same processing sequence. Even stability issues may be hard to realize, especially if the capacity of the lane is not large enough to store the order bins for all stores simultaneously. In this case, new order bins dynamically take over storage positions of already finished store orders and SKUs missed by the successors have to be moved along the lane again and again. This degrades picking performance considerably. Applications of put systems are mostly reported for fashion retailers such as the Reno (Nevada) distribution center of Urban Outfitters (2013).

Layout design: when setting up a put system two main decisions need to be taken: sizing the system holding capacity for order bins and dividing labor among multiple pickers. The former aspect is treated in Boysen, Stephan, and Füßler (2020); Füßler and Boysen (2017, 2019). The basic trade-off is to either enlarge the system holding capacity, so that more order bins can retrieve

the current SKU, which, however, increases the walking distance of the picker along the lane. Alternatively, a smaller capacity reduces each single walk along the lane of a picker, but increases the number of walks whenever a SKU is missed by another order bin not fitting into the lane. Füßler and Boysen (2017) compare different sizes of the picking lane and derive recommendations which size fits different order characteristics. They conclude that a rather small capacity should be chosen for larger assortments of SKUs and heterogeneous picking orders. For small assortments of SKUs and homogeneous picking orders, however, larger lane capacities are preferable. These results are confirmed by a subsequent study for a related warehousing system based on picking workstations, see Füßler and Boysen (2019). Boysen et al. (2020), instead, investigate put systems having enough capacity to store all order bins of all stores simultaneously. They benchmark different layout options (e.g., widely visible put-to-light displays, such that a picker can prematurely return whenever reaching the last order bin requiring the current SKU) and investigate their impact on picking performance. All these studies consider only a single picker operating the complete inverse order picking system. In most real-world systems, however, multiple pickers share the picking effort among each other. If the division of labor is organized, such that multiple pickers operate in overlapping areas along the lane, then picker blocking becomes an important aspect. Picker blocking arises whenever a specific order bin is currently filled by a picker and another picker aims access to the same bin at the same time (Parikh & Meller, 2009). Since an overtaking along the lane is typically not possible (i.e., this would require to lift the SKU bin from the belt for overtaking another bin), blocking also occurs due to different working speeds whenever a slower work impedes a faster one. Picker blocking due to different work speeds without overtaking has been considered for other warehousing systems, mostly traditional picker-to-parts warehouses with narrow aisles, e.g., (Chen, Wang, Qi, & Xie, 2013; Gue, Meller, & Skufca, 2006; Hong, Johnson, & Peters, 2012). However, the peculiarities of inverse order picking and whether overlapping areas shared among multiple pickers can indeed be synchronized with the put-to-light signals have not been considered in this context.

Storage assignment: the order bins placed along the put lane can be associated with facultative store orders, so that storage assignment in such a system is rather an operational problem, which can be altered with any new wave of store orders to be processed. Moreover, if a picker has to pass the complete lane anyway, then the storage assignment has no impact on picking performance. Boysen et al. (2020), however, consider the case where a picker only has to walk up to the last order bin requiring the respective SKU and can prematurely return. In this case, the storage assignment of order bins reduces the total waling distance of the picker and thus impacts picking performance. The authors suggest simple



Step 1) Order picking with vehicle support at ground level, e.g., steered by the pick-by-voice solution of tetes medea

Step 2) Replenishment from the upper reserve area, e.g., by a forklift of Linde Material Handling

Fig. 9. Picker-to-parts order picking with vehicle support.

priority rule-based approaches to determine suited storage assignments for this setting.

Order processing: when processing a given set of orders with a given storage assignment along the lane, the sequence in which the SKUs are presented down the lane may impact picking performance. This is the case, if the capacity for order bins is scarce and storage spaces freed by completed orders are taken over by new orders not yet processed. The resulting SKU sequencing and order batching problem, which decides on the succession of order bins taking over free storage space, is treated by Füßler and Boysen (2017) and Füßler and Boysen (2019). Both papers differ in the assumption whether or not a new order bin just having replaced a completed bin at a storage position can still receive the current SKU. If this is the case and also the sequence of succeeding store orders is given, the resulting optimization problem resembles the so-called sorting buffer problem, which is treated by Asahiro, Kawahara, and Miyano (2012) and Chan, Megow, Sitters, and van Stee (2012). Order processing in real-world systems with more than a single picker has not been investigated yet.

Given the widespread application of put systems, e.g., in the fashion industry, it is again surprising to see how few papers have addressed these systems so far. Future research should systematically record all different system setups of put systems (i.e., with and without enough capacity to store all order bins and with and without premature returns) and how to impact their performance via different storage assignments and SKU sequences.

8. Picker-to-parts systems with vehicle support

In picker-to-parts systems, human order pickers successively visit the storage positions of the SKUs defined on their pick lists and return to a central depot once their respective pick list is complete. Here, the completed order is handed over to the shipping area and a new pick list is started. Since store orders are large, i.e., they typically contain multiple order lines each demanding multiple pieces, the capacity of an unsupported human picker alone is often not large enough to handle a complete order in a single picker tour. Multiple returns to the depot with partial orders, however, considerably increase the fraction of unproductive

picker walking. Consequently, most warehouses servicing brick-and-mortar retail chains equip their pickers with motorized order pick vehicles (also denoted as *picking carts*) they can drive on (step 1 of Fig. 9). These motorized carts allow a faster travel through large warehouses and carry the pallets or roll cages for the current store order. To ease navigation through the warehouse and to reduce picking errors, nowadays, most pickers carry a pick-by-voice solution or a scanner system. Picking carts are typically not equipped with a lifting device, so that the picking area is restricted to the ground level. The upper shelf levels serve as the reserve area from where new unit loads are retrieved once the inventory at the ground level is (about to be) depleted. These replenishments are executed by extra forklift drivers (step 2 of Fig. 9), whose task is to keep the ground level supplied with SKUs without stockouts. To avoid interference among forklifts and picking carts, which typically may not pass a forklift in lifting mode for security reasons, many warehouse dedicate aisles to either picking or replenishing in an alternating manner. Some warehouses also apply a system of one-way picking aisles to further reduce interference.

The picker-to-parts system with vehicle support is the only warehousing system fulfilling all our seven requirements without exception. For instance, by adding additional picker-cart-tandems this system is flexibly scalable to workload peaks. Moreover, stability aspects and store-specific build-ups are enabled just by pickers processing each order in the desired sequence. Stability requires a unique processing sequence, which can easily be enabled by ordering the storage assignment according to these requirements (i.e., heavy and stable SKUs at the front close to the depot, light and fragile SKUs at the back). Store-specific build-ups, however, may vary from store to store, so that a unique storage assignment is not obtainable. Note that the level of variety may be lower in centrally controlled retail chains, which aim at standardized store layouts. Here too, however, differences in store ground plots and outlet sizes may cause at least some variation in store-specific build-ups. This variety may lead to pickers having to zigzag through the warehouse in order to retrieve the demanded pieces according to the individual sequence of the respective store. Thus, store-specific build-ups are possible, but may come for the price of excessive unproductive picker travel. Picker-to-parts systems are the most

basic warehousing setup, i.e., 80% of all warehouses in Western Europe are said to still follow this paradigm (De Koster et al., 2007; Napolitano, 2012), so that this setup is still very common among warehouses for all kinds of brick-and-mortar retail chains; it is the predominant system among all warehousing systems elaborated in this paper.

Picker-to-parts warehouses (with and without vehicle support) are not only very common in business practice, but also the most well-researched warehousing system with hundreds of research papers dealing with it. It is not our intention to review the complete research in this area and only refer to the excellent previous survey papers (De Koster et al., 2007; Gu et al., 2007; 2010; Masae et al., 2019; Rouwenhorst et al., 2000). Instead, we concentrate on the peculiarities added by servicing retail chains and applying motorized picking carts.

Layout design: planning the layout design in a picker-to-parts warehouse has to decide on the placement of racks, aisles and the depot within the warehouse. The decision support provided by existing research is, for instance, summarized by Rouwenhorst et al. (2000) and De Koster et al. (2007). The most common layout is the block layout where all racks are placed along parallel aisles with one or more cross aisles. In the recent years, quite a few papers have introduced unorthodox aisle designs to speed up the retrieval of picking orders. Among them are V-shaped aisles (Gue & Meller, 2009), the fishbone design (Pohl, Meller, & Gue, 2009), as well as the chevron, leaf, and butterfly designs (Öztürkoglu, Gue, & Meller, 2012). By applying analytical methods, these innovative layouts have been shown to improve picking performance compared to traditional block layouts. However, how to adapt these layouts if only every second aisle is available for the access of picking carts or safety reasons require one-way aisles has not yet been investigated. If the positive impact of these innovative layouts on picking performance is still existent with these adaptations, is an interesting question for future research. Moreover, the impact of one-way and dedicated replenishment aisles have also not been investigated for the traditional parallel aisle layout. The only paper specifically addressing a layout problem in a block layout when applying vehicle support is the one by Mowrey and Parikh (2014). They introduce a layout with different aisle widths within each of the parallel aisles. The basic intention is to let two oncoming vehicles pass each other in a wide aisle section, while utilizing the better usage of storage space in the narrow aisle sections. They suggest analytical methods where to place the wide aisle segments.

Storage assignment: for traditional picker-to-parts warehouses (without vehicle support) the storage assignment is certainly among the most well-researched topics of the warehousing domain. The manifold research efforts are summarized by the review papers of (Cormier & Gunn, 1992; De Koster et al., 2007; Gu et al., 2007; 2010). The most renowned storage policies are random storage, where each incoming unit load receives a new open, randomly selected storage position (e.g., see De Koster, van der Poort, & Wolters, 1999; Grosse, Glock, & Jaber, 2013; Tompkins, White, Bozer, & Tanchoco, 2010), dedicated storage, where each SKU receives a specific storage position (e.g., see Brynzér & Johansson, 1996; Frazelle, 2001), and class-based storage where SKUs and storage positions are prioritized into different classes and only suited matches are allowed (e.g., see Jarvis & McDowell, 1991; Petersen & Schmenner, 1999). How the manifold results on the performance of these storage policies are altered, if only every other one-way aisle is dedicated to picking, has not been considered yet. Moreover, if the racks are separated into a bottommost picking area and the uppermost replenishment area, then each new incoming unit load should receive a storage position in the replenishment area close to the current unit load of the same SKU placed in the picking area. Then, the replenishment effort for the forklift drivers can be reduced, once the unit load in the pick-

ing area needs to be replaced. How to achieve this, especially if many unit loads of the same SKU arrive over time, still waits to be explored. While stability issues can simply be considered by storing the heavy and stable products closer to the depot, so that they can be placed onto the bottommost layers of pallets and roll cages at the start of each picker tour (Dekker, De Koster, Roodbergen, & Van Kalleveen, 2004; Žulj et al., 2018), deriving storage assignments suited for store-specific build-ups is not that straightforward. Each store has its own specific layout and, thus, demands another sequence in which the products are placed on the pallets and roll cages. Given the store layouts and their demand probabilities for all SKUs, new storage assignment rules are required, which allow a suited compromise if one store suggests to store a SKU closer to the depot and another rather prefers the opposite.

Order processing: in a picker-to-parts warehouse and its large fraction of unproductive picker travel, the operational routing problem has outstanding impact on throughput performance. With vehicle support, two kinds of entities need to be routed: (i) the forklifts executing the replenishment and (ii) the pickers driving their picking carts during order picking.

(i) Given a set of replenishment moves for a single forklift, the former routing problem, integrating the horizontal movement of the vehicle and the simultaneous vertical movement of the crane, constitutes a special kind of traveling salesman problem (TSP) where all moves start or end at the bottom most picking area. Boysen and Stephan (2016) have shown that, in spite of the special structure of the distance matrix, the resulting TSP is still strongly NP-hard, even if all replenishment moves are located in the same aisle. For the same special case, Kim et al. (2003) provide a simple sorting based heuristic and a more efficient clustering-based algorithm.

(ii) The routing of a single picker in a warehouse (i.e., with a given set of picking positions where pieces are to be retrieved, in a parallel aisle warehouse with cross aisles at the front and back) also constitutes a special case of the TSP. In their seminal paper Ratliff and Rosenthal (1983) have shown that this special case is solvable to optimality in polynomial time with dynamic programming. Their approach has successfully been extended by Roodbergen and De Koster (2001) as well as De Koster and van der Poort (1998) if an additional middle cross aisle is available and the handover of picked products is enabled in the front and back cross aisle via conveyors, respectively. Only recently Pansart, Catusse, and Cambazard (2018) have generalized this approach to multi-block warehouses. They apply the polynomial-time algorithm of Cambazard and Catusse (2018) for the rectilinear Steiner tree problem in the plane to solve picker routing with a facultative number of cross aisles to optimality. Furthermore, many heuristic solutions, such as the S-shape and largest-gap heuristic, which are said to be more intuitive for pickers, have been explored (De Koster et al., 2007). After this brief general overview, we will focus those aspects relevant when routing a picker driving on a picking cart:

- **Cart stops:** getting on and off the cart may take the picker some time, so that it may be advantageous to retrieve multiple SKUs from close-by storage position during a single cart stop. In this case, the stop time of the cart has to be traded off against the picker's lower walking velocity when retrieving products on foot. As an additional restriction, the limited handling capacity of a picker when carrying multiple pieces has to be considered. An exact dynamic programming approach for resolving this trade-off within a single aisle has been presented by Goetschalckx and Ratliff (1988). They compare their approach with a simple heuristic clustering method and conclude that considerable savings can be realized.
- **Blocking:** picking carts carrying a pallet or roll cage for larger store orders are bulky and require considerable space, so that

they may block each other within picking aisles. By introducing one-way systems, it can be avoided that two oncoming carts oppose each other in an aisle and one of them has to back out of the aisle. However, it is still possible that a cart parked for picking blocks or delays another cart with the same driving direction. Integrating blocking into picker routing is very challenging, because the problem no longer decomposes into single-picker problems. [Chen et al. \(2013\)](#) consider blocking of two pickers in a narrow aisle warehouse. They introduce an ant colony optimization algorithm and benchmark their approach with a straightforward extension of the S-shape heuristic. [Chabot, Coelho, Renaud, and Côté \(2018\)](#) investigate a case study of a warehouse for furniture and electronic equipment and treat the resulting picker routing problem with blocking. They model the problem as a vehicle routing problem by using a series of distance transformations between all pairs of locations and show that it can be decomposed into a set of single-aisle problems. The problem is tackled by branch-and-cut and large neighborhood search. A warehouse layout with only a single cross aisle at the front, so that each aisle can only be accessed from a single opening is considered by [Polten and Emde \(2019\)](#). They differentiate two different access policies: exclusive and parallel aisle access. In the former, only a single vehicle at a time is allowed per aisle, whereas the latter allows access of multiple picking carts, which have to be coordinated within the aisle. MIPs and NP-hardness proofs for both policies are provided, and the problem is solved by large neighborhood search. Additionally, to explore different storage assignments and picking policies under picker blocking on a higher planning level, analytical methods have been introduced by [Parikh and Meller \(2009\)](#) and [Gue et al. \(2006\)](#). [Franzke, Grosse, Glock, and Elbert \(2017\)](#) apply an agent-based simulation study to explore the impact of storage assignment combined with a specific routing policy on picker blocking. Another simulation study is provided by [Van Gils, Caris, Ramaekers, Braekers, and De Koster \(2019\)](#). They show that applying traditional warehouse routing methods (designed for single pickers and wide aisles) can lead to a considerable loss of throughput, if applied in a real-world setting where safety issues and picker blocking lead to interaction effects of multiple pickers.

- **Precedence constraints:** if stability aspects need to be considered or store-specific build-ups have to be realized, the interdependencies among the SKUs on a pick list can be translated into precedence constraints, defining that one storage position has to be accessed before the other. [Matusiak, De Koster, Kroon, and Saarinen \(2014\)](#) introduce a joint order-batching and picker routing method to solve the combined precedence-constrained routing and order-batching problem. The authors present a MIP formulation based on a variant of the precedence-constrained TSP. For solving their problem, they decompose the problem: An exact A*-algorithm is applied for solving the picker routing problem and a fast simulated annealing algorithm for order batching. [Chabot, Lahyani, Coelho, and Renaud \(2017\)](#) integrate weight, fragility and category constraints into picker routing. They present two different MIPs, a capacity-indexed and a two-indexed network-flow formulation. The problem is tackled with exact branch-and-cut methods and a heuristic large neighborhood search algorithm. Another solution approach for precedence-constrained order picking is presented by [Žulj et al. \(2018\)](#). The authors extend the approach of ([Ratliff & Rosenthal, 1983](#)) by defining different types of sub-tours to incorporate precedence constraints. For a case study, they show that the total handling effort can considerably be reduced compared to a system where SKUs are traditionally picked (without precedence constraints) and sorted afterwards in the depot.

It can be concluded that, although quite a few papers treating picker-to-part systems with vehicle support already exist, there is still room for future research. Especially, storage assignment policies for store-specific build-ups are a challenging field not yet explored.

9. Future research needs and conclusion

[Table 1](#) summarizes the papers for each warehousing system and decision problem. Note that we only count those papers specifically solving a decision problem for the warehousing system, and not all papers we refer to in the respective section (e.g., on related problems or survey papers). Further note that papers may occur multiple times of they treat multiple problems. This summary table reveals that many areas are barely investigated. Especially the storage assignment problems in all non-standard warehousing systems not based on the conventional picker-to-parts or the pick-and-pass setup are rarely addressed. Note, however, that bulk picking is rather a process adaption with a consolidation conveyor as a constituting element. Here, the lack of storage assignment research does not indicate a massive research gap, because conventional storage assignment techniques can be applied here, see [Section 6](#). Moreover, the peculiarities of bag sorters, although being a frequently selected consolidation technology in many recent greenfield warehouses, have not been considered in the scientific literature yet. The same holds true for the store-specific build-ups elaborated in [Section 2](#) and how to efficiently realize them. This leaves plenty of room for future research.

However, we also see some major trends and special features of brick-and-mortar retail chains beyond our previous discussions on alternative warehousing systems that should be addressed by future research.

- **Parallel warehousing systems:** we see a clear trend toward applying multiple warehousing systems in a single distribution center each dedicated to different SKUs and/or alternative order characteristics. Note that this can be seen as an adaption of the renowned strategy-focused factory concept of Wickham Skinner, see e.g., [Brumme, Simonovich, Skinner, & Wassenhove, 2015](#), which contains the advice of “dividing the facility into areas each with a different focus”. For instance, a warehouse could combine a bulk picking system with store consolidation (see [Section 6](#)) for its most fast-moving SKUs in combination with a picker-to-parts system with vehicle support (see [Section 8](#)) for the wide range of slow movers. In this way, a suited warehousing system for different operational needs is available. The price, however, is additional investment costs and the need to consolidate partial orders for the same store from multiple warehousing systems. This trend is triggered by omni-channel retailing where retailers aim to give customers access to their products both via online and offline sales channels ([Agatz et al., 2008](#)). Distribution centers having to realize omni-channel demands face the low-volume-high-mix picking orders typical for e-commerce, but also high-volume-low-mix demands of brick-and-mortar stores. Another reason for applying multiple warehousing systems in parallel are varying workloads. During end-of-season sales most distribution centers face much higher demands, so that less flexible automated systems (e.g., fully-automated case picking, see [Section 3](#)) for the base volume can be combined with manual systems (e.g., picker-to-parts system with vehicle support, see [Section 8](#)), which are flexibly adjustable to peak loads. Scientific decision support when to apply multiple warehousing systems and which combinations fit what branches of industry are yet missing.
- **Omni-channel warehousing:** in their empirical study on excellence in omni-channel retailing [Hübner et al. \(2016a\)](#) conclude

Table 1
Papers per warehousing system and decision problem.

Warehousing system	Layout design	Storage assignment	order processing
Fully-automated case picking	–	(Chen et al., 2010)	(Bischoff, 2006; Crainic et al., 2009; Faroe et al., 2003; Junqueira et al., 2012; Kang et al., 2012; Lodi et al., 2002b; Martello et al., 2000)
Pick-and-pass systems for large orders	(Armbruster et al., 2007; Bartholdi & Eisenstein, 1996; Bartholdi et al., 2001; Jane, 2000; Jane & Laih, 2005; Jewkes et al., 2004; Koo, 2009; Melacini et al., 2011; Van der Gaast et al., 2018; Otto et al., 2017; Yu & De Koster, 2008)	(Füßler et al., 2019b; Jane, 2000; Jane & Laih, 2005; Jewkes et al., 2004; Van der Gaast et al., 2019; Otto et al., 2017; Pan et al., 2015; Pan & Wu, 2009; Yu & De Koster, 2008)	(Füßler et al., 2019a)
Crane-supplied pick face	(Ramtin & Pazour, 2014)	(Ramtin & Pazour, 2015)	(Boysen & Stephan, 2016; Kim et al., 2003; Schwerdfeger & Boysen, 2017; Yu & De Koster, 2010)
Bulk picking and store consolidation	(Bozer & Sharp, 1989; Johnson & Lofgren, 1994; Russell & Meller, 2003)	–	(Boysen et al., 2018b; Bozer et al., 1988; Gallien & Weber, 2010; Johnson & Lofgren, 1994; Kizilaslan et al., 2013; Petersen, 2000)
Put systems	(Boysen et al., 2020; Füßler & Boysen, 2017; 2019)	(Boysen et al., 2020)	(Asahiro et al., 2012; Chan et al., 2012; Füßler & Boysen, 2017; 2019)
Picker-to-parts systems with vehicle support	(Mowrey & Parikh, 2014)	(Dekker et al., 2004; Žulj et al., 2018)	(Chabot et al., 2018; Chabot et al., 2017; Chen et al., 2013; Franzke et al., 2017; Goetschalckx & Ratliff, 1988; Gue et al., 2006; Matusiak et al., 2014; Parikh & Meller, 2009; Polten & Emde, 2019; Van Gils et al., 2019; Žulj et al., 2018)

that about half of their interviewed non-food retailers propagate integrated warehouses when servicing both online and offline orders. On the positive side, integrated warehouses allow to pool inventory and bundle transports, e.g., when delivering click-and-collect orders of customer households jointly with regular replenishment orders to brick-and-mortar stores (Hübner et al., 2015). The price, however, is a loss in picking efficiency when handling high-volume-low-mix store orders and low-volume-high-mix online orders jointly via the same warehousing system (Kembro et al., 2018). A common extension policy we observed quite often when a traditional brick-and-mortar retail chain using picker-to-parts warehouses (see Section 8) aims to integrate online orders into its existing processes goes like this: the online orders are batched to large picking orders, so that their structure resembles store orders. After picking, an additional sort-after-pick consolidation is required, which (either manually or via a consolidation conveyor, see Section 6) separates customer orders. These process adaptations and the related costs are one important argument for the other half of the sampled retailers of (Hübner et al., 2016a) to rather opt for separate structures, which can either be realized by opening additional channel-specific distribution centers or by outsourcing to third-party logistics providers. A compromise strategy in between both extremes is to apply multiple parallel warehousing systems in the same distribution center (see above). Which of these three alternatives on how to realize omni-channel warehousing is the appropriate choice in a specific case certainly depends on the extent in which the structures of online and store orders deviate. Many small shops are supplied on a daily basis via a postal service provider, so that their demands rather resemble online orders (Hübner et al., 2015). On the other hand, if large restaurants order their food demand online at a food wholesaler, these orders can exceed those of small mom-and-pop stores. More research (both empirical and analytical) on which order structures of online and store orders suggest to either integrate, separate, or mix warehousing systems is much in demand to support the widely pursued transformation to omni-channel retailing. This finding is

supported by the recent structured literature review of Kembro et al. (2018) on omni-channel warehousing. They report a lot of literature on rather qualitative and/or strategic aspects in this domain, but see a lack of operations research approaches addressing specific decision problems. Thus, we believe that omni-channel retailing and its impact on warehousing systems will be one of the most important research issues in the years to come.

- *Competitive advantages:* from a logistics point of view, brick-and-mortar retail chains have three main structural advantages over e-commerce retailers: (i) store-based retail chains suffer much less from products returns. Most online retailers face extraordinary return rates up to 40% (Morganti, Seidel, Blanquart, Dablanc, & Lenz, 2014). Products returned to distribution centers of retail chains are mostly defect and can directly be claimed against the respective suppliers, because returns due to displeasure or wrong size reenter the forward chain directly in the respective store (De Koster et al., 2002). (ii) Brick-and-mortar stores have continuous demands triggered by customer sales, which are much easier to forecast. This allows to bundle demands to larger orders, so that transportation costs are reduced and transport can be synchronized with warehousing operations (Sternbeck & Kuhn, 2014). (iii) More predictable demands allow to channel products ordered from suppliers directly to the stores without storing them in a distribution center. Thus, many retailers make extensive use of cross-docking, which reduces their storage costs (Apte & Viswanathan, 2000; Boysen & Fliegener, 2010). Traditional retail chains should systematically exploit these three advantages over e-commerce to build up a competitive advantage. Cost savings realized on the logistics side can then be passed on to the customers, e.g., in terms of lower prices or better services. A specific measure on the warehousing side could, for instance, be the application of a bag or pouch sorter system (see Fig. 7). The better forecast quality in retail chains eases the question, which pieces to store forward in the bags, and cross docking requires a consolidation device anyway. However, more scientific decision support on how these structural advantages can systematically

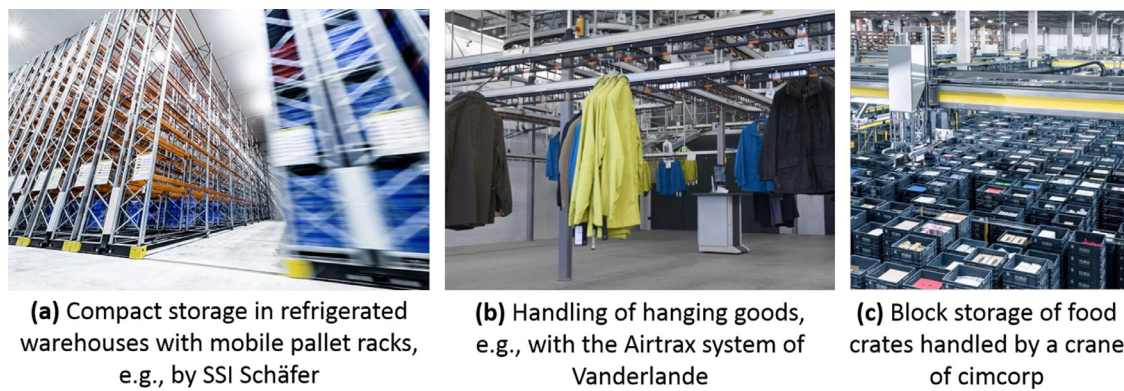


Fig. 10. Special warehousing systems for specific retail branches.

be exploited when setting up and operating warehouses is required.

- *Human factor*: the aging workforce in many industrialized countries suggests to integrate ergonomic aspects into workplace design (Otto et al., 2017). Moreover, a cooperation of robots and human workers becomes a tangible alternative, also in the warehousing domain (e.g., Magazino, 2017). The survey papers of Grosse, Glock, Jaber, and Neumann (2015); Grosse, Glock, and Neumann (2017) on human factors in warehousing, however, reveal that not much research in this direction exists. Specifically, the ergonomic stress of excessive walking could be addressed when applying pick-and-pass systems (see Section 4) and put systems (see Section 7). Just as well, the reduction of ergonomic strain of the sales personnel when not having to zigzag heavy roll cages through the stores due to store-specific build-ups (see Section 2) deserves further scientific attention.
- *Holistic research*: to allow for a rigorous analysis, the vast majority of existing papers focus on an isolated problem of a single subsystem of a warehouse (e.g., an ASRS). Today's warehouses, however, are complex facilities, so that, from the warehouse managers' perspective, holistic models are desirable that take a rather unifying look at the complete order fulfillment process. Bulk picking, for instance, requires a delicate coordination of multiple stages, i.e., picking, induction and consolidation, to avoid idle times and gridlocks (see Section 6). Due to their great interdependence these stages cannot be analyzed in an isolated manner. Satisfying the demand for practical, holistic models will remain a huge challenge for future research. This aspect is specifically emphasized by the recent survey paper of Van Gils et al. (2018) where combinations of multiple planning problems are focused.
- *Branch specifics*: although excluded from our survey (see Section 2), there exist interesting warehousing systems for specific retail branches. They are challenging, yet mostly neglected fields for future research too and, to give an example, we briefly sketch three of these systems: (i) Especially in refrigerated warehouses, e.g., for frozen food, considerable savings in cooling costs of more than 30% can be realized by high-density storage (Material Handling Industry of America (2009)). One possibility to realize a more compact usage of space are mobile racks (see Fig. 10(a)). Here, parallel racks are mounted on rails, such that a strong engine can move them sideways and open an aisle to access the SKUs (Boysen et al., 2017; Foughi, Boysen, Emde, & Schneider, 2020). There also exist quite a few alternative high-density storage systems, which are surveyed in detail by Azadeh et al. (2019). All these compact storage systems trade off efficient space usage against slower retrieval times, so that organizing these systems such that an acceptable throughput is enabled, constitutes a challenging field of research. (ii) Cloth-

ing is often handled by automated sorting systems for hanging goods (Schäfer, 2018a). Here, goods-on-hangers, e.g., for dresses and coats, are stored on slick rail lanes and can automatically be sorted into other lanes assigned to a specific store (see Fig. 10(b)). There is flexibility on how to channel the SKUs via a system of switches, so that optimization procedures steering this process are an interesting, yet neglected research field. (iii) Finally, some food retailers sort crates of fruit and vegetables according to store demands via crane systems lifting one or multiple crates from block storage (see Fig. 10(c)). Similar systems are applied to handle tires (Cimcorp, 2011), and these systems also resemble a block storage of containers in ports. A famous decision problem in the latter field optimizes the retrieval sequence of containers stacked on top of each other and is called the blocks relocation problem (e.g., see Lehnfeld & Knust, 2014). Warehousing cranes, however, can lift multiple crates stacked on top of each other simultaneously. Containers are much heavier, so that typically only one container at a time can be lifted. Special systems being able to lift more than a single container are investigated by Zhang, Liu, and Kopfer (2016). However, whether this latter approach can directly be transferred to the warehousing domain is yet to be explored. On a more aggregate level, a general overview which warehousing systems especially suit what retail branches with their specific products, requirements, and order characteristics would be valuable decision support for practitioners having to select a suited warehousing system. An attempt in this direction is, for instance, provided by Yoon and Sharp (1996), who provide a structured selection procedure how to identify and design a suited warehousing system. Some empirical evidence on branch specifics among 68 Italian warehouses is provided by Dallari et al. (2009). More research in this direction is definitely required.

It can be concluded that organizing the fulfillment processes of brick-and-mortar retail chains is an important topic with many open research challenges. Seeing the fierce competition of e-commerce and the strong pressure to enable efficient warehousing processes it seems fair to project that order fulfillment will remain a lively field of research in the years to come.

CRediT authorship contribution statement

Nils Boysen: Conceptualization, Methodology, Investigation, Writing - review & editing. **René de Koster**: Conceptualization, Methodology, Investigation, Writing - review & editing. **David Fülßler**: Conceptualization, Methodology, Investigation, Writing - review & editing.

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